

Assignment 01: Site Modeling with LiDAR and Physical Fabrication

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Course: Arch 565 – Advanced Computer Applications II
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Overview

This assignment challenged me to blend new digital documentation tools with classic physical modeling approaches to build a hybrid site model. I captured part of a built environment and its topography using publicly available LiDAR-derived data, processed it into a precise 3D model, and prepared it for physical fabrication. The aim was to build expertise in spatial precision, digital-to-physical transitions, and thoughtful analysis of computational tools in architecture.

Objectives Met

- Hands-on work with LiDAR-based site documentation.
- Process point cloud or contour data into practical 3D models for architectural review.
- Merge digital modeling and fabrication to create a physical site model.
- Assess the precision, drawbacks, and potential of LiDAR compared to traditional site methods.

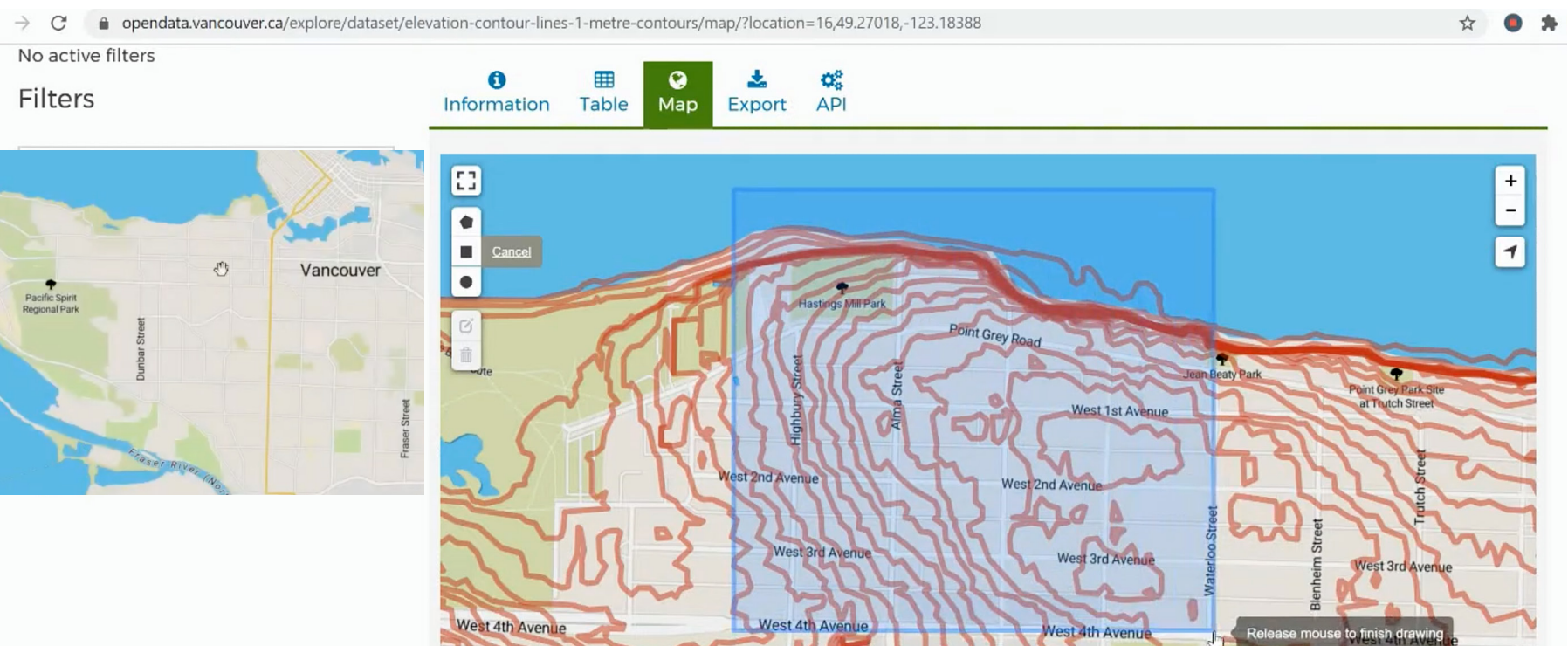
Site Selection & Scanning

I gathered LiDAR-derived data for the project site from publicly available sources, specifically Vancouver’s open data portal. The focus area was a small urban block in Kitsilano along West 4th Avenue, near Pacific Spirit Regional Park — a site characterized by subtle slopes and dense building footprints, making it well-suited for testing topographic accuracy.

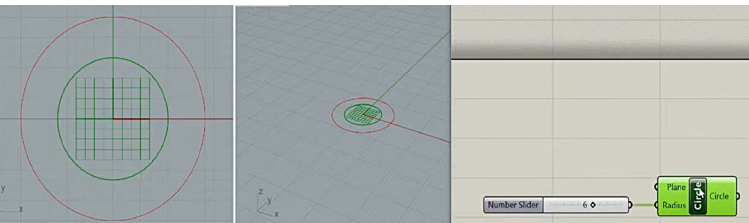
To obtain the contour data, I navigated to opendata.vancouver.ca and searched for “elevation contour lines 1-metre contours.” I selected the dataset with 1-meter intervals, previewed it on the map, and drew a bounding rectangle around the site to export only the relevant portion. I ensured the export used the EPSG:26910 coordinate system to preserve geolocation accuracy, then downloaded the selected records as a shapefile (.shp). On Windows, I unblocked the file through its properties before extracting it.

For the building data, I searched for “building footprints 2009” and reviewed the attribute table, which included details such as elevations and roof types. I defined a similar bounding area around the site and exported the data in the same EPSG:26910 format as a shapefile.

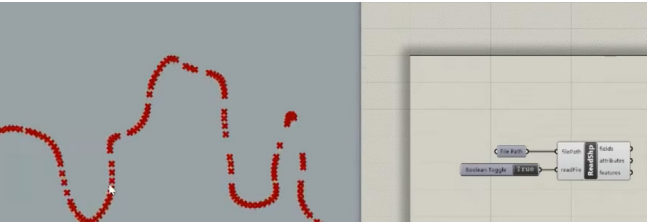
Open Data Vancouver



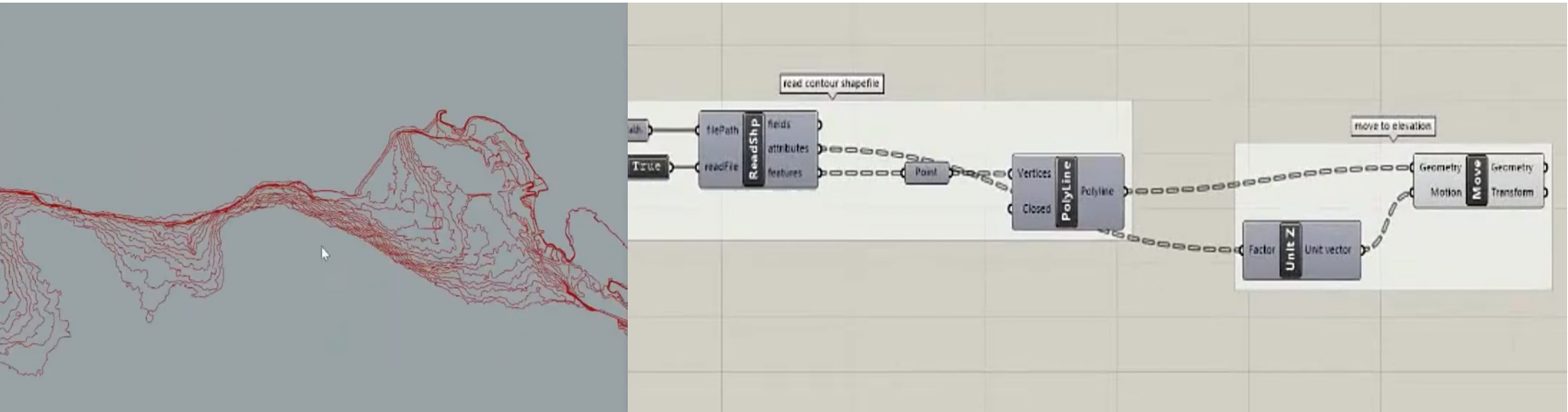
Making The plane



Just Points



To Poly Lines and Moving to Elevation



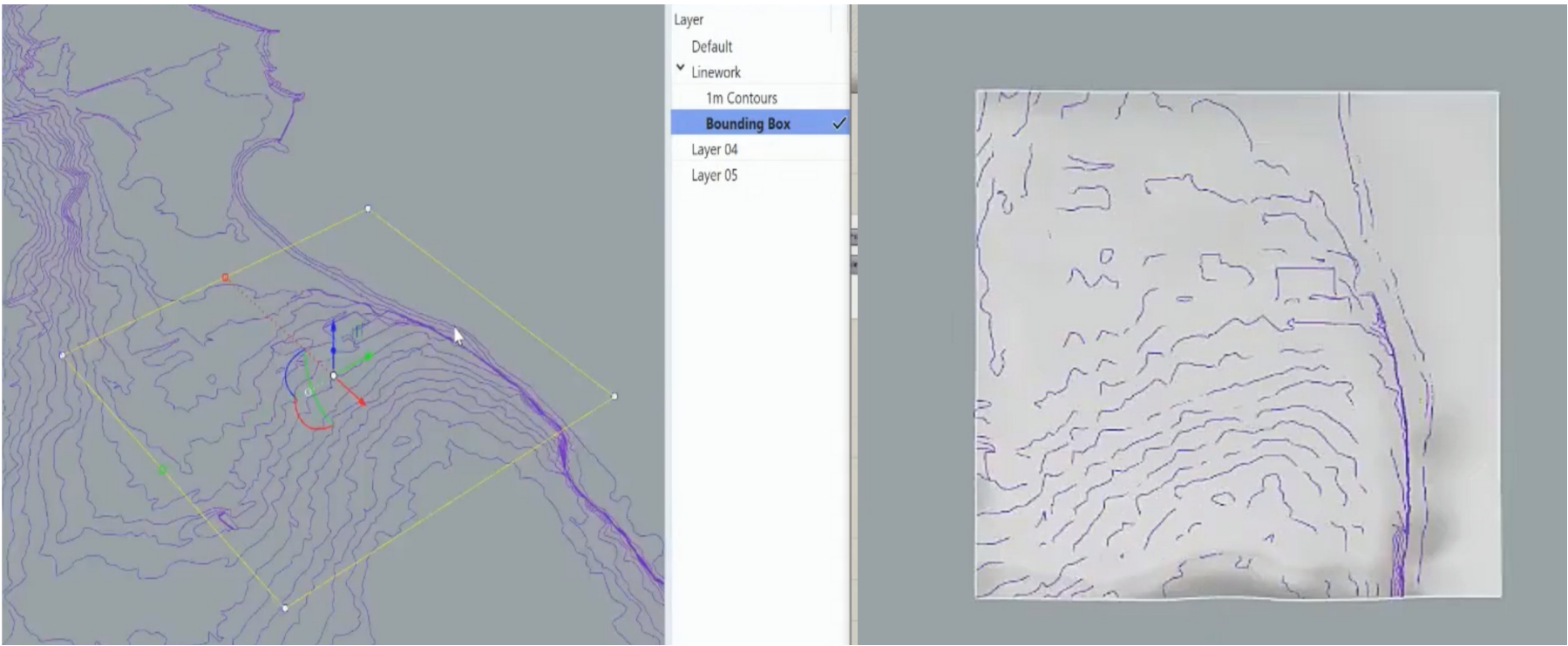
Trimming Date to fit the 50' x 50' Size

Data Processing & Digital Modeling

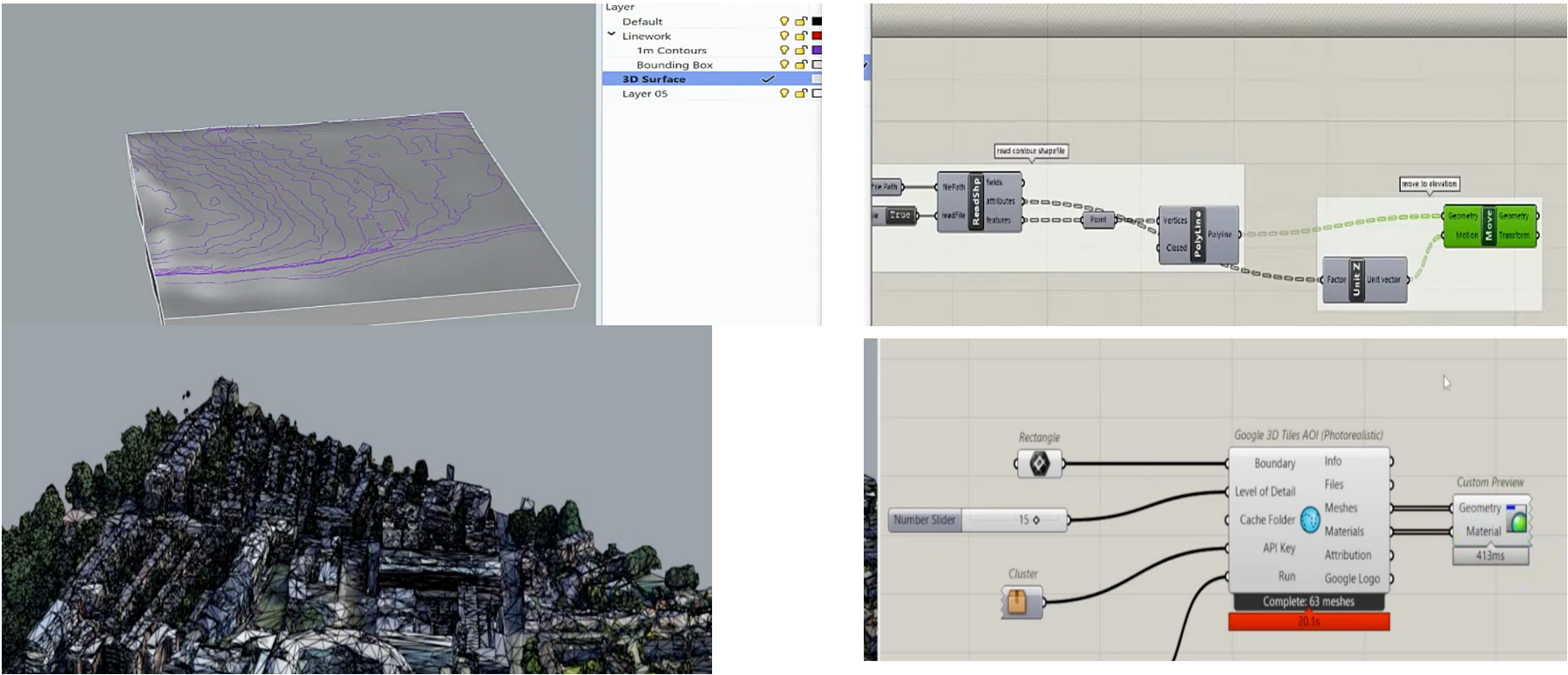
I imported the data into Rhino and Grasshopper using plugins such as Heron for contours and Elk (or equivalent) for buildings.

For contours, I placed a “Read Shape” component on the canvas, assigned a file path to the .shp file, and connected a boolean toggle set to true. This generated a set of points, which I then connected into polylines using a “Polyline” component with the points as vertices. To assign elevation, I used a “Move” component with the polylines as geometry and a “Unit Z” vector as the motion direction, scaled by the elevation attributes extracted from the features via a “List Item” component at index 0 (or the appropriate index for elevation).

I organized these steps into two groups — “Read Contour Shape” and “Move to Elevation” — then baked the polylines onto a “1-Meter Contours” layer in Rhino. I created a bounding rectangle on a “Linework” layer, trimmed any excess contours, and generated a surface using the “Patch” command applied to the elevated contours, with U/V spans set to 30 and stiffness at 5 to balance accuracy with processing time. The surface was then extruded downward to form a base, trimmed to a clean edge, and capped.



Patching and Fill

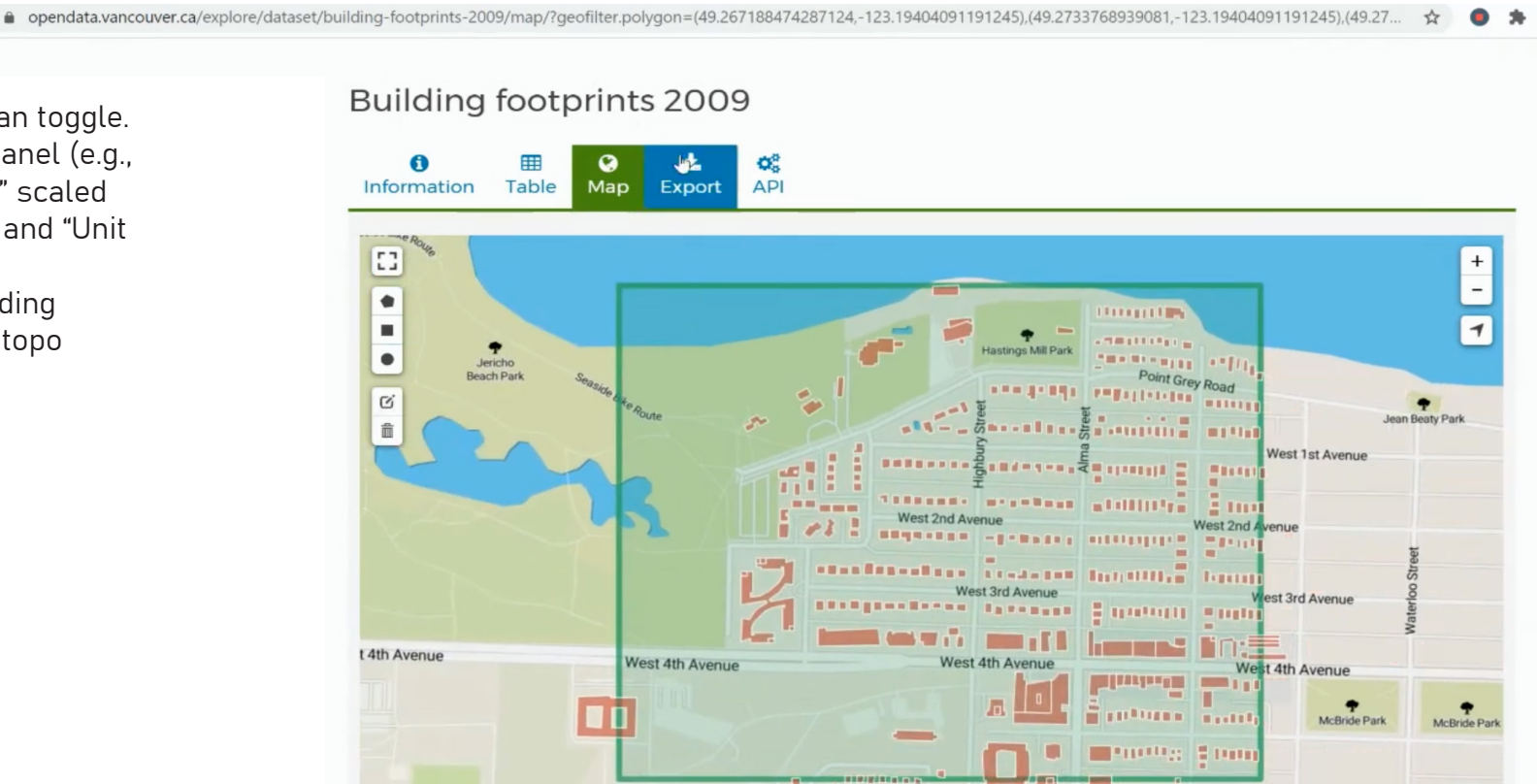


Using point clouds and meshes to fit the surface then extruding points and filling to solid. And Utalizing 3D Tiles for mapping date.

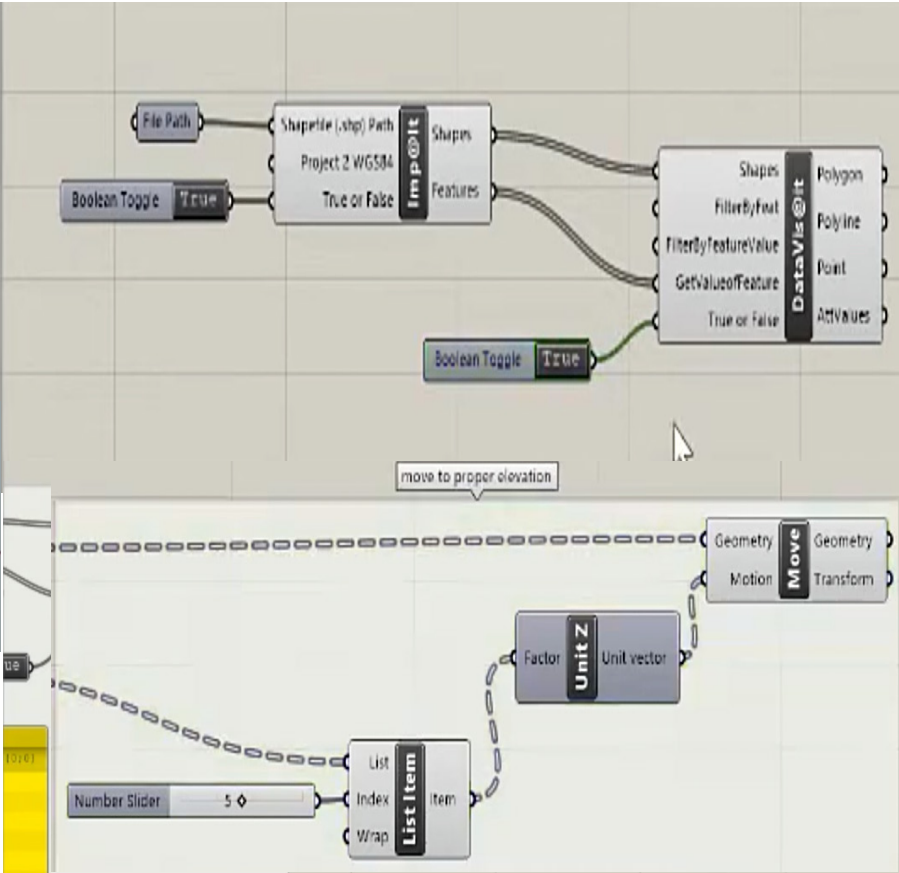
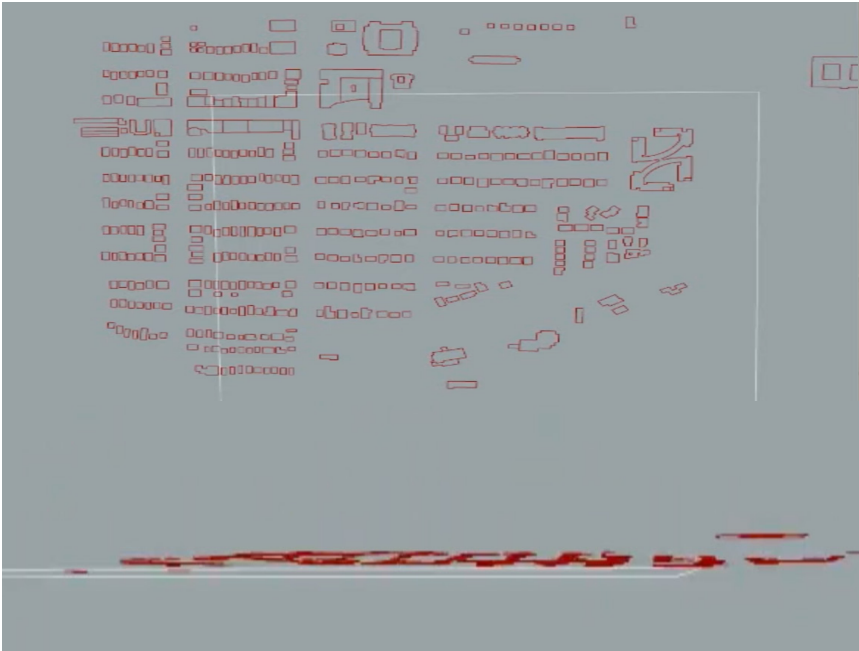
Getting The Building Data

Buildings

For buildings: I used an “ImpEdit” component with a file path to the building .shp and a boolean toggle. I visualized data with “DataViz,” connecting shapes and features, and viewed attributes in a panel (e.g., base elevation at index 5, max height at index 11). I moved polygons using “Move” with “Unit Z” scaled by base elevation via “List Item” at index 5. I extruded using “Extrude” with polygons as base and “Unit Z” scaled by height via “List Item” at index 11, then capped with “Cap Holes.” I grouped these as “Import Building Shapefile,” “Move to Proper Elevation,” and “Extrude Building Heights.” I baked to a “3D Buildings” layer, trimmed partial buildings, and integrated with the topo surface for a complete model.

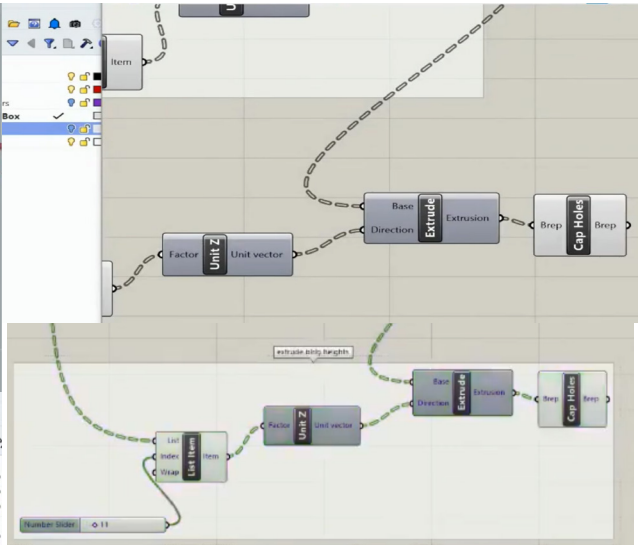
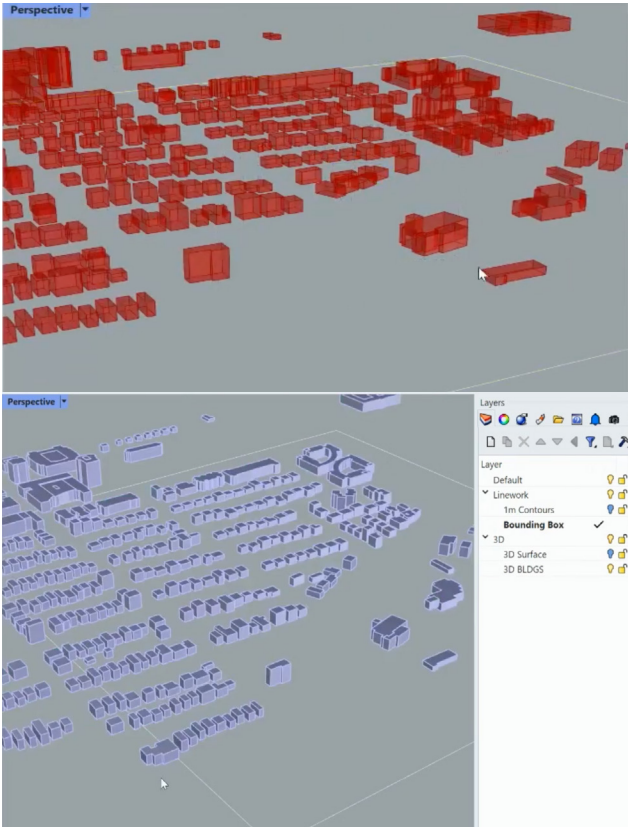


Dataviz: List and Index



Setting the buildings to the respective elevations.

To Solid

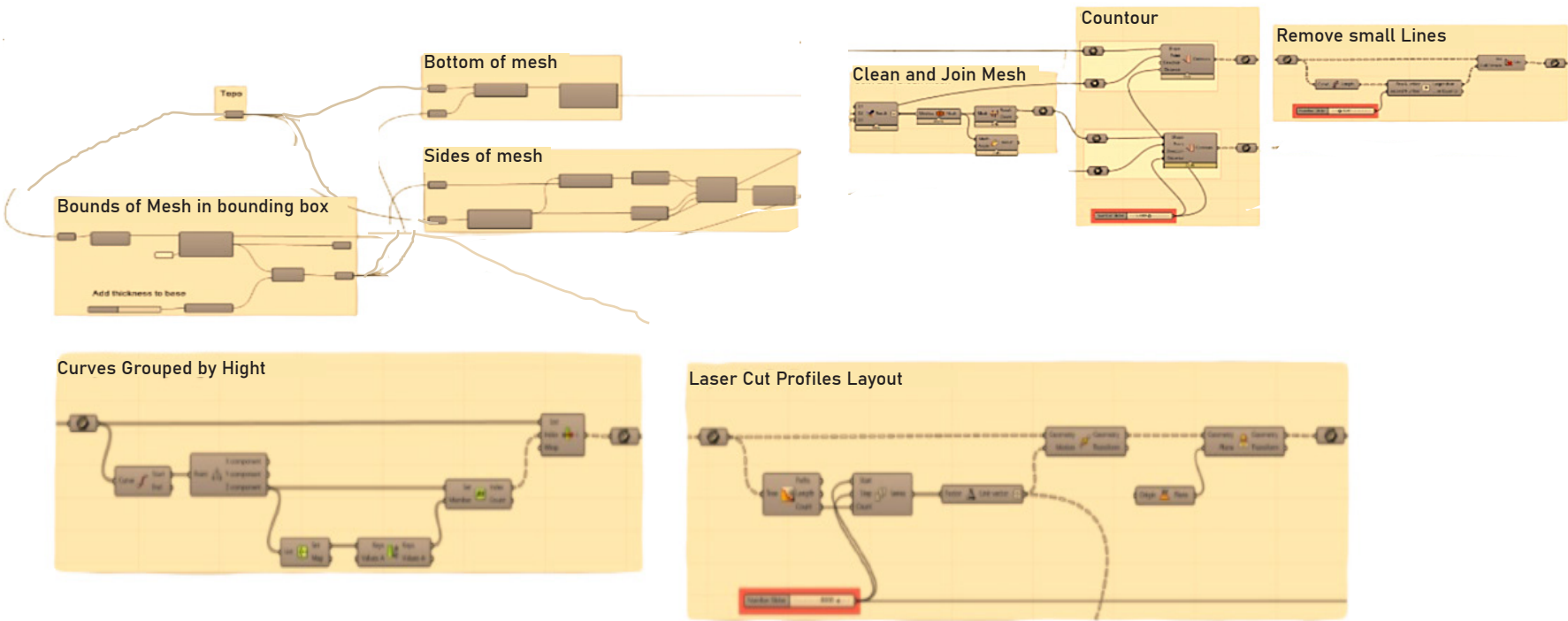


Capping the solid geometry then baking

Ready For Laser Cutting

Physical Fabrication

I prepared the digital model for fabrication by creating a clean, boxed mesh suitable for laser cutting. I bounded the mesh with a rectangle, trimmed edges for 90-degree corners, and adjusted thickness for stability. I formed the base and walls using “Project” for the bottom and meshed connections between top and bottom. I joined faces with “Mesh Join,” cleaned with “Weld Mesh” and “Unify Mesh,” then contoured for laser-cut layers. I translated to fabrication files (.dwg for laser cutting), chose a scale of 1:200, and assembled the physical model with layered cardstock, ensuring clear representation of topography, buildings, and features.



Laser Cut Bed

Documentation & Reflection

This report documents the workflow with images illustrating key steps, from data export through final model assembly. LiDAR-derived data from public portals provided high accuracy for urban sites, capturing subtle 1-meter elevation changes and building details with efficiency. Processing in Grasshopper streamlined elevation adjustments and extrusions, significantly reducing manual effort compared to traditional surveying methods.

However, data gaps in building footprints necessitated manual trimming, and patch-generated surfaces occasionally deviated from the contour lines, revealing limitations when working in dense urban environments. Overall, this technology accelerates the iteration process but requires verification against on-site observations. It complements hand-built models by enabling precise hybrid approaches, creating opportunities for sophisticated site analyses in professional practice.

